

Are Turkish neighborhoods becoming politically more uniform?

What five parliamentary elections' worth of ballot boxes — about 919,000 of them — say about political sorting inside Turkey's mahalles, 2011–2023.

1. The question

A few years ago, Turkish researchers did something laborious and clever. They wanted to know how a gated community voted — specifically, whether the people inside the walls voted differently from the people outside. Official results don't tell you that: the finest thing published is the sandık, the individual ballot box, and nothing in the data says which side of a wall a box sits on. So they interviewed local party organizations to learn which sandık numbers belonged to the gated complex, and found what they suspected: boxes a wall apart voted sharply differently. One neighborhood on paper; two political worlds on the ground.

That study covered one complex, by hand. The obvious question is what happens when you compute the same thing for the whole country, across time: **are like-minded voters increasingly concentrated together — are neighborhoods becoming internally more homogeneous, or more divided?** This is the Turkish version of what Americans call the "Big Sort" question, and for Turkey the nationwide, longitudinal version had not been measured.

This paper is the answer we can currently defend. The short version:

Within neighborhoods that stayed comparable across all five elections from 2011 to 2023, the vote shares of the three long-established parties — AKP, CHP, and MHP — became measurably **less** dispersed across the ballot boxes of the same neighborhood. Turkish mahalles, on this measure, became modestly more internally uniform, with the clearest decline for CHP.

The long version is mostly about how easy it is to get this wrong, and what it took to make the sentence above survive. That story — three traps found, one seductive finding killed — is at least as interesting as the headline.

2. The data

The raw material is the box-level results of the five parliamentary (milletvekili) general elections: June 2011, June 2015, November 2015, June 2018, and May 2023. After cleaning, each election contributes between roughly 174,000 and 199,000 usable ballot boxes — about 919,000 box-level records in all, covering electorates growing from 46 million registered voters in 2011 to 56 million in 2023.

Each box record carries its geography: province (il), district (ilçe), neighborhood (mahalle), and box number. The mahalle is the key. A mahalle typically contains a handful to a few dozen boxes, each serving a few hundred registered voters assigned by address. So while we never see an individual's vote, we do see — at address-block resolution — how uniformly or unevenly a party's support is spread inside each neighborhood. That within-mahalle granularity is what mahalle-aggregated studies cannot compute, and it is the distinctive asset here.

Two honest limits of the data, up front:

- **For 2011, 2018, and 2023, the mahalle is the finest geographic key.** Only the two 2015 elections carry school/building names. So boxes within a mahalle can be compared to each other, but not placed on a map more finely than their mahalle.
- **Everything here is areal.** We measure boxes and neighborhoods, not people. Nothing in this paper says any voter changed their mind, their party, or their address.

Cleaning this data is its own small war, and two details deserve mention because they silently corrupt results if ignored. First, Turkish text: the dotted and dotless *İ*/*ı* do not survive standard lowercasing, so a naive `lower()` turns *İSTANBUL* and *ISPARTA* into keys that no longer match their own records — every place-name join runs through a Turkish-aware normalizer, and this is enforced by tests. (The same bug bit this project's own chart labels during figure production, which says

something about how easy it is to reintroduce.) Second, vote-share denominators: shares are computed against reported valid votes (geçerli oy), with an explicit residual column for small parties and independents, rather than against the sum of major-party columns — the latter silently inflates shares in exactly the spatially correlated way that would fake a sorting trend. The 2018 and 2023 alliance-era files keep alliance lump votes in separate, clearly labeled columns, never folded into party totals.

3. The measure

For each mahalle, each election, and each party, we ask: how much do this party's vote shares vary across the ballot boxes of this one neighborhood?

Concretely: take each box's party share (party votes ÷ valid votes), and compute the spread across the mahalle's boxes — a standard deviation, weighted by box size. A mahalle where AKP polls 40% at every box has spread zero: a politically uniform neighborhood, whatever its lean. A mahalle where AKP polls 20% at half the boxes and 60% at the other half is internally divided — the gated-community pattern, visible without knowing where the wall is.

Raw spread has a flaw for trend purposes: it scales with the party's level of support. A party sliding from 45% to 10% will mechanically show smaller absolute spread even if its geography is unchanged. So the workhorse statistic is the **coefficient of variation (CV)**: the spread divided by the party's mean share in that mahalle. CV asks "how dispersed is this party's support, relative to its own size here?" — which is the question we actually mean. National summaries weight mahalles by registered voters, so the series describes where voters actually live rather than treating a 300-voter village and a 40,000-voter urban mahalle as equal observations.

4. Three ways to fool yourself, and the ladder built against them

The naive computation — CV per mahalle, national average per election, draw the line — produces a satisfying decline. The project treated that line as a suspect, not a finding, and checked three artifact channels.

Trap 1: the crowded field. The party system fragmented between 2011 and 2023; more parties means lower shares, which distorts raw spreads. Using CV rather than raw SD addresses the mechanical part, and the decline survives: comparing only the parties present throughout (AKP, CHP, MHP), the voter-weighted national CV falls from 2011 to 2023 for all three (AKP 0.165→0.134, CHP 0.325→0.177, MHP 0.351→0.284, uncontrolled).

Trap 2: box counts. The number of boxes per mahalle changed over time, and measured dispersion correlates with box count (more boxes, more room to disagree — partly a small-sample effect). This correlation was material in 11 party-election cells, so box-count stability became an explicit control condition rather than a footnote.

Trap 3: boundary churn — the big one. Mahalle identifiers do not simply carry over between elections. Looking up each mahalle of one election by name in the next: from 2011 to June 2015, only 42% of mahalle names can be re-found (the 2012–2014 metropolitan-municipality reform re-labeled huge numbers of villages as mahalles; the data records the churn, not its cause). From 2015 onward, names mostly match (~85–87%) — but the matched names hold only about half the registered voters, meaning the unmatched names are on average the much more populous places. A "trend" computed on whatever matches year to year would silently swap large parts of the electorate in and out and call the result change.

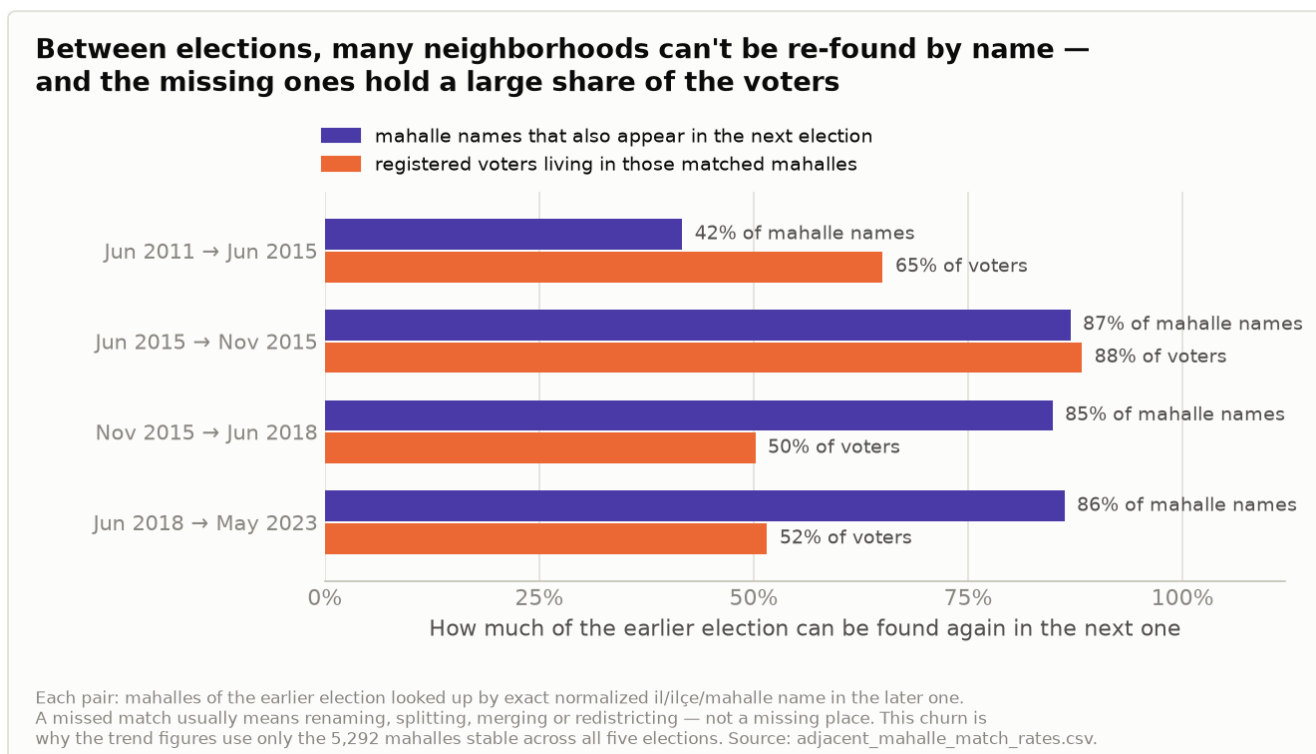


Figure 5. Adjacent-election continuity: share of mahalle names re-found in the next election (violet) and of registered voters living in the matched mahalles (orange).

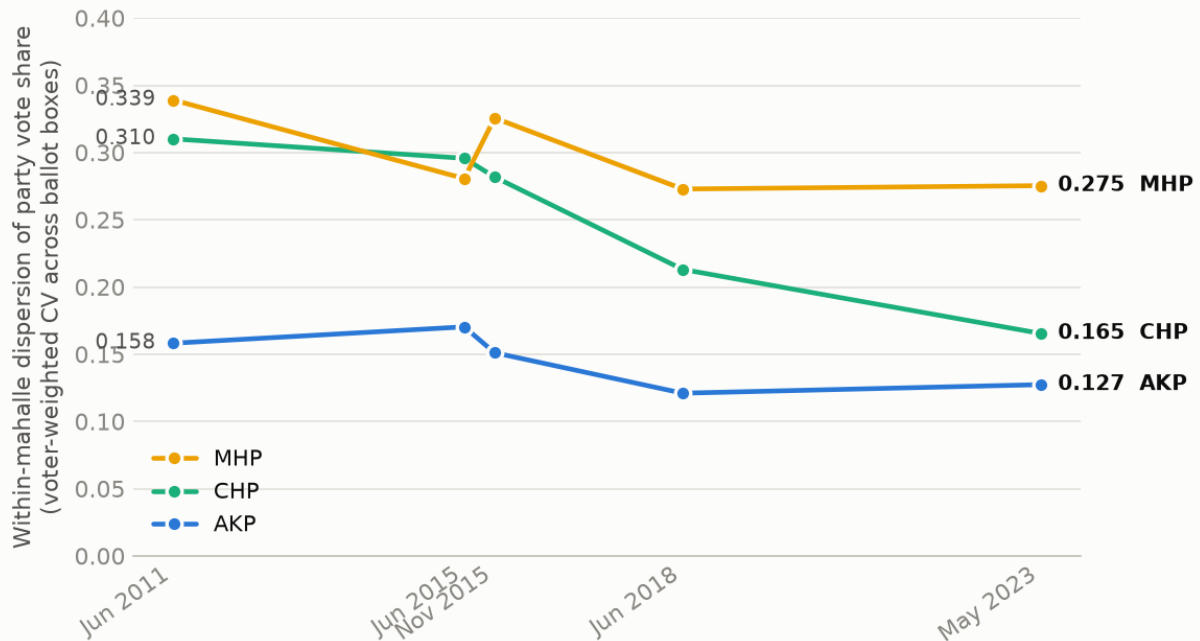
The response is a ladder of increasingly strict conditions, ending in **condition 4**: only mahalles whose province/district/mahalle key exists in all five elections, and whose box count stays stable across the whole run (at least 2 boxes every time; max/min ratio ≤ 1.5 , or an absolute range of at most 2). That leaves **5,292 full-chain stable mahalles**, covering about 9.5 million registered voters in 2011 and 11.3 million in 2023 — roughly a fifth of the electorate, deliberately traded for comparability. A province-anchored version of the statistic serves as an independent sanity check on direction.

5. The finding

On the strict condition-4 subset, voter-weighted within-mahalle CV:

PARTY	2011	2023	DIRECTION
AKP	0.158	0.127	decline
CHP	0.310	0.165	strong decline
MHP	0.339	0.275	decline

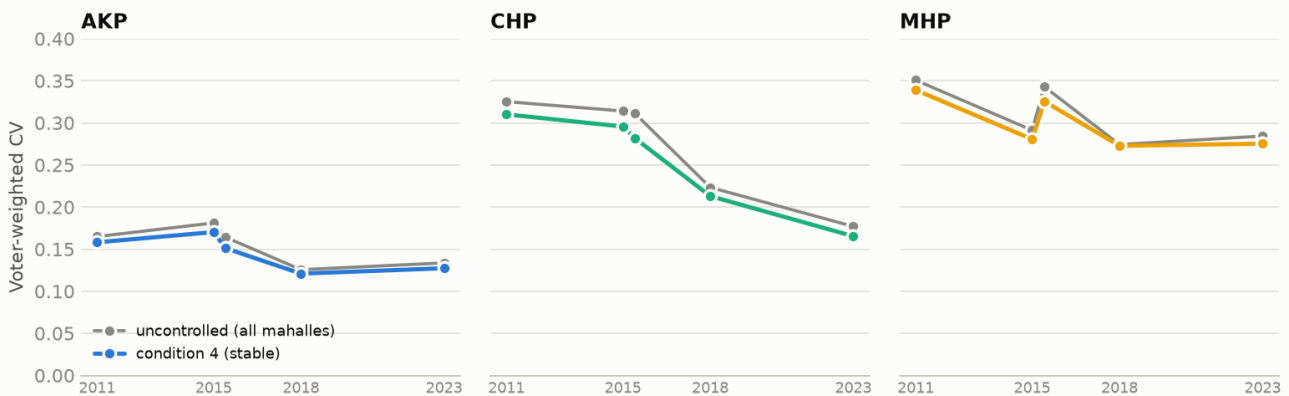
Within stable mahalles, AKP, CHP and MHP vote shares became less dispersed across ballot boxes, 2011-2023



Strict control (condition 4): 5,292 full-chain stable mahalles with stable box counts, 9.5M (2011) to 11.3M (2023) registered voters. Areal measure over ballot boxes; says nothing about individual voters. Source: m3_four_condition_trends.csv, voter-weighted.

Figure 1. Within-mahalle dispersion (voter-weighted CV) for AKP, CHP, MHP on the 5,292 full-chain stable mahalles of condition 4.

The decline is not a boundary-churn artifact: it survives the strict stable-geography control



Gray: all mahalles, no controls. Color: condition 4 — full-chain stable mahalle keys and stable box counts. Source: m3_four_condition_trends.csv.

Figure 3. Uncontrolled national series (gray) versus condition-4 stable-geography series (color). The decline survives the strictest control.

Three things about this result.

It survives the controls, visibly but not trivially. The uncontrolled and condition-4 series differ — the control is doing real work — but the direction holds in both, for all three parties. The decline is not an artifact of boundary churn or shifting box counts.

The metric is not a one-way ratchet. MHP's raw within-mahalle spread rises from 2011 to June 2015 (0.033→0.034) before falling in later elections. A pure mechanical-compression artifact would not let a party move against the tide; that it can is quiet evidence the measure captures party-specific geography.

The magnitude is modest, and we say so. CHP's within-mahalle CV roughly halved over twelve years; AKP's and MHP's declined by about a fifth. This is a real, consistent compression of within-neighborhood political variation among the established parties — not a transformation of the country. "Turkey homogenized" is not the claim, and the data as analyzed so far would not support it.

6. The finding that wasn't: how the HDP/YSP artifact was caught

For a while, the analysis appeared to contain a second, more dramatic story. The Kurdish-movement lineage (HDP in 2015–2018, running as Yeşil Sol/YSP in 2023 — a well-documented continuity that the party-code crosswalk handles explicitly) showed the opposite trend: within-mahalle CV **rising** from 0.524 in June 2015 to 0.630 in 2023 on the same strict stable subset. Rising internal fragmentation for one party, against homogenization for the big three, in a politically charged corner of the data — an irresistible headline.

It is wrong, and the way it is wrong is instructive. A coefficient of variation explodes where a party's support approaches zero: with a 1% mean share, trivial absolute differences between boxes become enormous relative spread. If the apparent rise is real neighborhood-level fragmentation, it should survive restricting to mahalles where the party actually has support. It does not — it reverses:

MAHALLES INCLUDED (CONDITION-4 SUBSET)	JUN 2015 CV	2023 CV	DIRECTION
All	0.524	0.630	rises
Party $\geq$ 5% support	0.341	0.243	declines
Party $\geq$ 10% support	0.268	0.168	declines
Party $\geq$ 20% support	0.157	0.101	declines

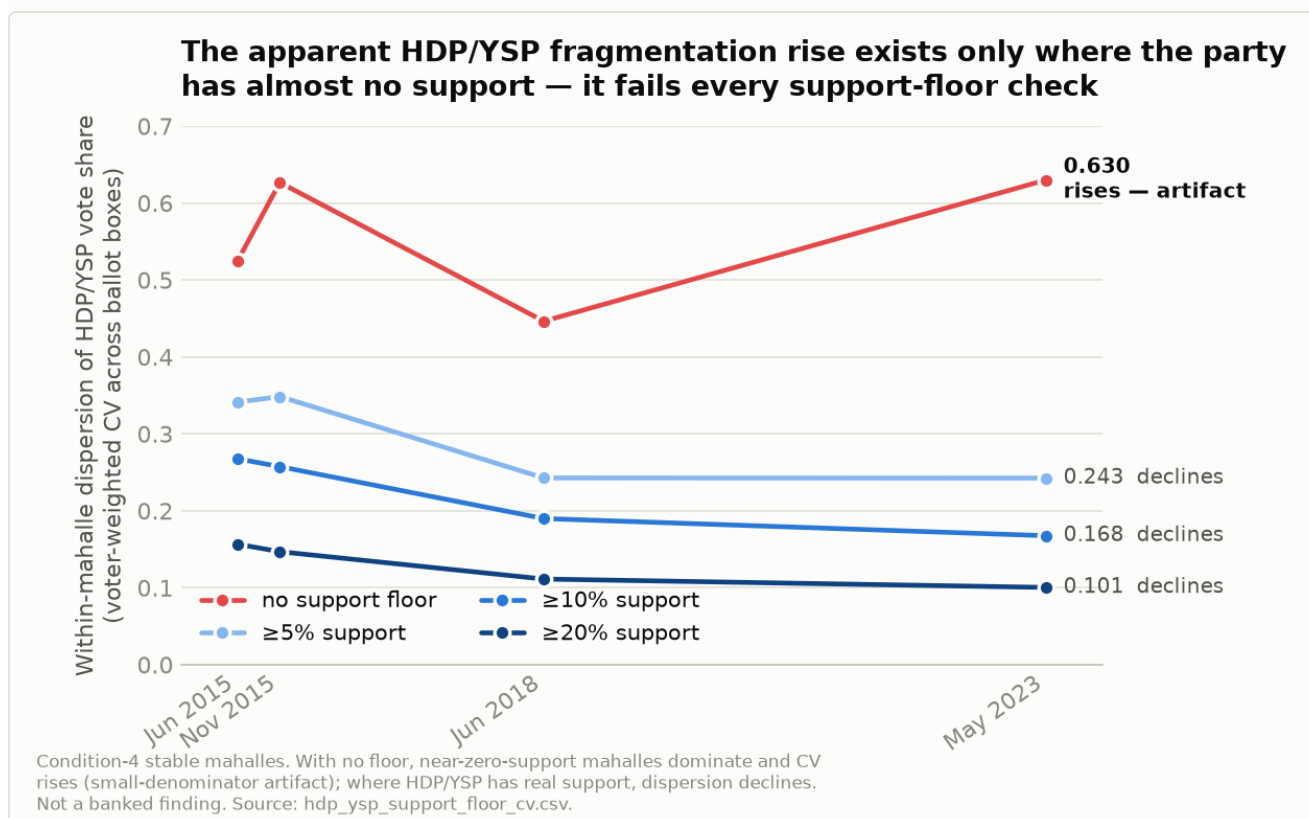


Figure 2. HDP/YSP within-mahalle CV on the condition-4 subset, with and without support floors. The rise exists only with no floor — a small-denominator artifact, not a finding.

Wherever HDP/YSP has a real presence, its neighborhoods homogenized like everyone else's. The nationwide "rise" lives entirely in near-zero-support mahalles — a small-denominator artifact, not sorting. Two corroborating checks: the independent province-anchored series never showed the rise (roughly flat-to-down while the strict mahalle series rose — the divergence that triggered scrutiny), and turnout-composition checks in the southeast found no strong turnout-driven explanation to rescue a substantive reading.

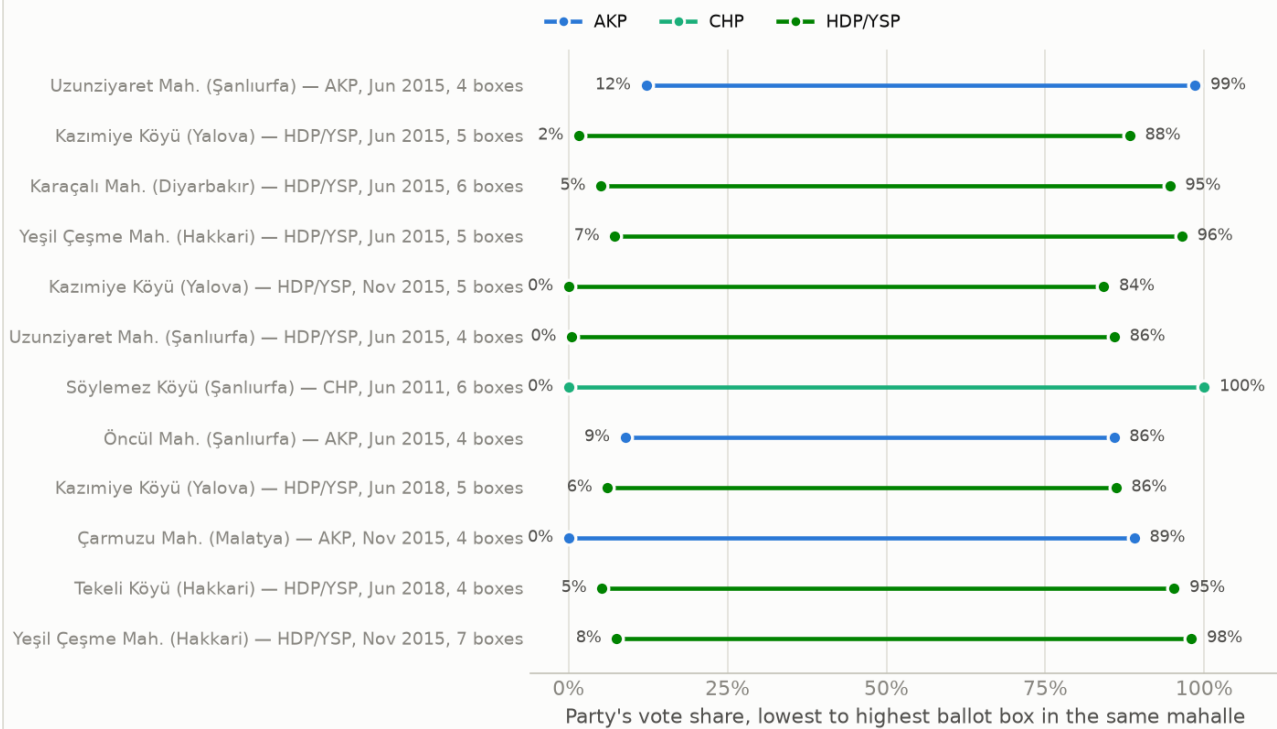
So the fragmentation claim is dead, and this paper does not make it. It was the most tempting result in the project, which is exactly why it got the most hostile examination.

7. Neighborhoods split down the middle

The national trend is one use of the data. The other is the automated version of the gated-community study: rank every party-mahalle-election cell by internal spread and look at the extremes.

To keep tiny-box noise out, the flagship list requires at least 4 boxes and at least 150 registered voters per box; the filtered top-100 overlaps the unfiltered one by 81%, so the extremes are not merely small-denominator noise. The cases are striking. In Uzunziyaret Mahallesi (Siverek, Şanlıurfa) in June 2015, AKP polled 12% at one ballot box and 99% at another — four boxes, one neighborhood, 885 registered voters. In Kazımiye Köyü (Yalova), HDP ranged from 2% to 88% across five boxes — and the same place reappears in the extreme list in November 2015 and 2018. Boxes are filled by address, so splits like these are shadows of real social boundaries — a street, a settlement pattern, perhaps a wall — visible in public data without anyone conducting an interview.

Extreme within-mahalle splits: the same party and neighborhood, very different boxes



Top 12 party-mahalle-election cells by within-mahalle share SD, after noise filters (≥ 4 boxes, ≥ 150 registered voters per box). Local micro-fragmentation examples, not national-trend evidence. Source: top_fragmented_filtered.csv.

Figure 4. The most internally split party-mahalle-election cells after noise filters (≥ 4 boxes, ≥ 150 registered voters per box): lowest and highest box share within one neighborhood.

These are local portraits, not trend evidence: a list of extremes is by construction unrepresentative. Their role is to make concrete what "within- mahalle dispersion" means, and to flag candidate sites for local inspection.

8. A map, honestly labeled

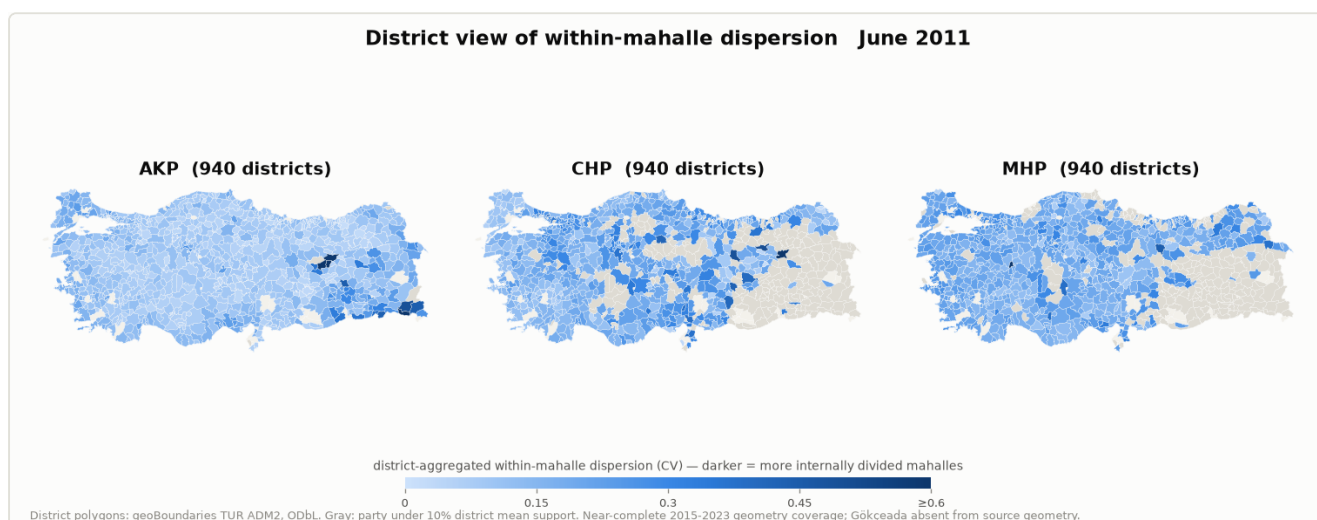


Figure 6a. District aggregation of within-mahalle dispersion per party, June 2011 (first frame of the animation). Gray = under 10% district mean support or no party list.

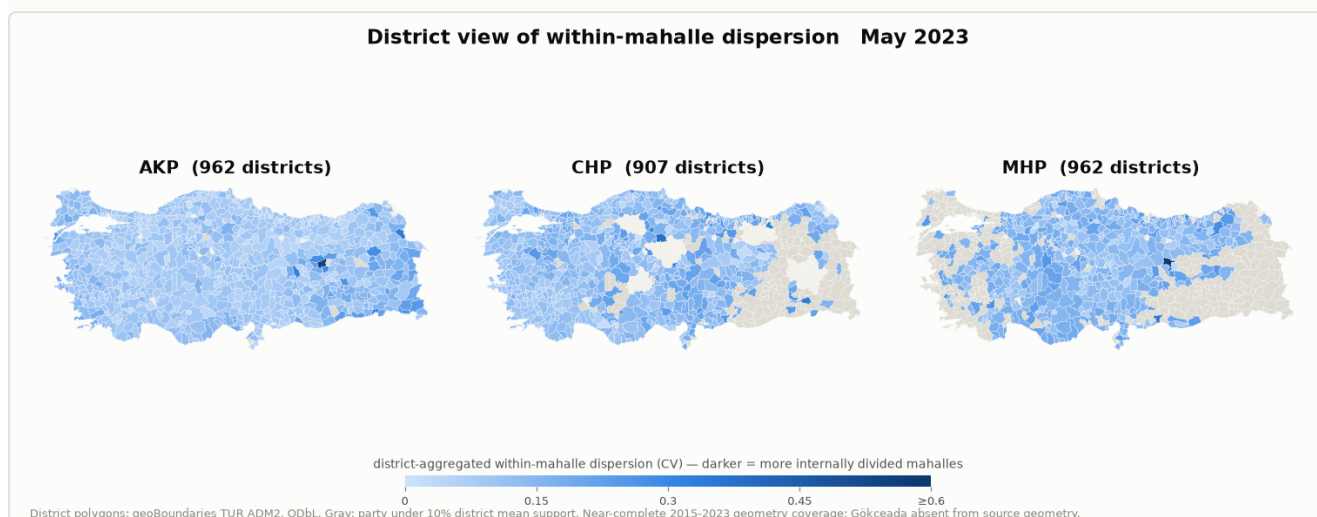


Figure 6b. The same district maps, May 2023 (last frame). The animated version steps through all five elections.

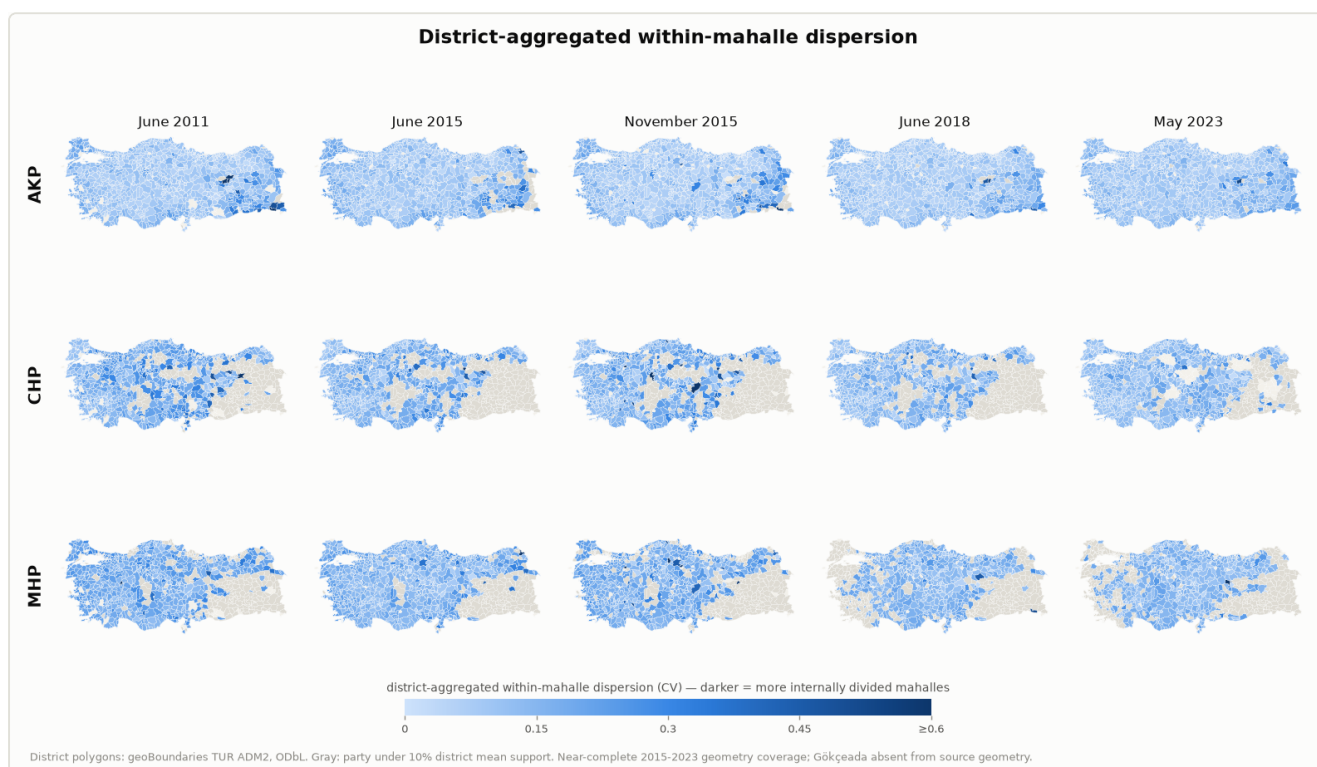


Figure 6c. All five elections as a static district grid — rows fade toward lighter (more internally uniform mahalles within districts).

The map everyone wants — every mahalle shaded by its internal division, animated over five elections — requires national mahalle boundary polygons, which still do not exist in this project (see §9). The most granular defensible filled map now available is district-level: each ilçe polygon is shaded by the voter-weighted within-mahalle dispersion of the mahalles inside that district, per party, per election. The district geometry comes from geoBoundaries' Turkey ADM2 polygons; for 2015–2023 the join is effectively complete (only Gökçeada is missing from the source geometry in the current elections), while 2011 is shown with a stronger historical-boundary caveat.

At this resolution the finding is visible in a more local way than the province map: the CHP panels lighten distinctly from 2011 to 2023, AKP remains mostly low with pockets of higher within-mahalle dispersion, and MHP fades more modestly. Districts are masked gray where a party polls under 10% (the same small-denominator problem as §6 — CV can look dark when the party's mean share is small, and painting those cases would reintroduce the exact artifact we killed), and where a party has no usable list/geometry for that panel. This support-floor caveat is part of Figure 6's interpretation, not just a web-map interface detail.

The map is a descriptive companion to the trend result, computed as district aggregations of mahalle-level dispersion; it illustrates the finding, it does not independently prove it.

We also asked the natural next spatial question: if boxes inside stable mahalles became more uniform, did stable mahalles also form stronger geographic clusters with nearby mahalles? Using OSM place-node centroids for the condition-4 stable subset, longitudinal FDR-screened LISA transition tests do not show a robust 2011-to-2023 change in spatial clustering for AKP, CHP, or MHP. AKP shows endpoint transition structure under one tested weight rule ($k=8$) but not under the primary, $k=6$, or 10 km-cap rules; CHP and MHP do not show robust start-end transition signals. The coverage check makes this a clean exploratory null rather than a data-sparsity artifact: the centroid join covers 4,253 of 5,292 stable mahalle keys, 85.3% of 2023 registered voters in the stable subset, and about 90.7% of southeast voters in that subset.

So the main result should not be promoted into a broad claim that Turkey became more geographically clustered at mahalle-centroid resolution. The defensible claim is narrower: established-party support became less dispersed inside stable mahalles, and the stronger between-mahalle clustering version of the story does not survive the longitudinal LISA robustness gate. HDP/YSP remains even more fragile in this spatial layer, because its fine-resolution results depend on support floors and spatial-weight choices.

9. What this does and does not establish

The caveats are part of the result, not disclaimers around it.

- **Areal, not individual.** This is sorting of ballot boxes within neighborhoods. Whether it reflects residential moves, registration changes, local conversion, generational turnover, or something else is not identified. No claim about individual behavior is made or implied.
- **Modest magnitude.** Falling within-mahalle CV for three established parties, strongest for CHP. Not "Turkey transformed."
- **The stable subset is not a random sample.** Condition 4's 5,292 mahalles are places whose names and box counts stayed put — plausibly

more settled and more rural than the churning metropolitan periphery the matching excludes. The trade of coverage for comparability is explicit, and the uncontrolled series (which agrees in direction, on ~19,000–24,000 mahalles per election) is always shown alongside.

- **Stable controls match mahalles, not boxes.** Within a stable mahalle, individual boxes are still renumbered and reshuffled between elections; the panel is at neighborhood grain, not box grain.
- **Resolution limits are unresolved (MAUP).** All dispersion is measured within administrative mahalle units. The 2015 elections' school names would allow a finer-resolution recomputation to test how much the unit choice matters; that calibration has not been done.
- **This extends prior work; it does not found a field.** Spatial autocorrelation studies of Turkish voting exist at province and district level (mostly single-election snapshots), one mahalle-level GIS study covers a single province (Hatay, 2018), district-divergence work builds on the Johnston et al. spatial-polarization tradition, and the gated-community study supplied the box-level idea. The contribution here is doing it nationwide, longitudinally, at box-within-mahalle grain, with a matched stable-geography panel — systematizing at scale what existed in pieces.

10. Where this could go next

Three natural extensions, none required for the result above: geocoding the 2015 school names to calibrate how much the mahalle aggregation costs and to build a nationwide "wall detector" for that year; adding richer local context for the extreme split-mahalle cases; and using registration/turnout dynamics to probe mechanism — whether homogenization travels with population churn or happens among stable populations. Each is a genuinely harder project than what is banked here.

Appendix: provenance

Every number in this paper traces to a file in the analysis repository, and every figure to a script:

- Cleaned data: `outputs/cleaned_mv/` (five elections; conformance reports alongside). Cleaning contract and tests: `CLEANING_SPEC.md` , `tests/test_cleaning_contract.py` (passing, 4/4, verified this session).
- Condition-4 trend and coverage (§5, Figs 1 & 3): `outputs/m3_boundary_boxcount_control/m3_four_condition_trends.csv` , `m3_condition_coverage.csv` ; control definitions in `m3_audit.json` .
- Uncontrolled CV check and box-count correlations (§4): `outputs/pre_m3_artifact_checks/artifact_check_verdict.json` .
- HDP/YSP resolution (§6, Fig 2): `outputs/hdp_ysp_resolution/hdp_ysp_support_floor_cv.csv` , `hdp_ysp_verdict.json` .
- Fragmented-mahalle list (§7, Fig 4): `outputs/pre_m3_artifact_checks/top_fragmented_filtered.csv` .
- Continuity diagnostics (§4, Fig 5): `outputs/cleaning_audits/adjacent_mahalle_match_rates.csv` .
- Map (§8, Fig 6): `outputs/geo_join/district_map_values.csv` , geometry from local `geoBoundaries-TUR-ADM2-all.zip` , join audit `outputs/geo_join/geoboundaries_adm2_join_coverage.csv` , rendered by `scripts/make_district_choropleth_map.py` .
- M2 longitudinal LISA null (§8): `docs/m2_lisa_transitions_status.md` , `outputs/m2_lisa_transitions/` , `outputs/m2_closeout_checks/` , `scripts/m2_lisa_transitions.py` , and `scripts/m2_closeout_checks.py` .
- Figures: `scripts/make_findings_figures.py` , `scripts/make_district_choropleth_map.py` , and `scripts/make_homogenization_map.py` → `outputs/figures/` .

Findings register: `FINDINGS_BANK.md` . No claim in this paper is stronger than its entry there.